

Manipal Hackathon 2026



Team Name : CtrlFreaks

Track : SMART GOVERNANCE & COMPLIANCE

Problem Statement : Automated Public Procurement Anomaly Detection

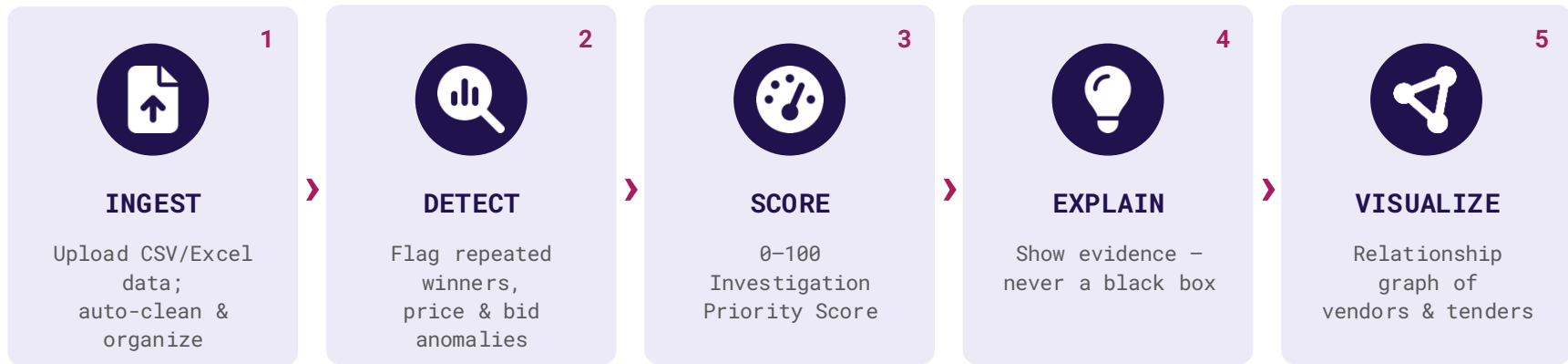
Solution Title : Procurement Intelligence & Investigation Radar

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SOLUTION



An AI-assisted procurement auditing platform that analyzes tenders, bids, vendors, contracts and payments.



Humans stay in the loop: the system flags cases for investigation – it never declares anyone corrupt.

TECHNICAL IMPLEMENTATION



FRONTEND



React / Next.js dashboard
with charts & graph view

INTELLIGENCE



Investigation
Engine
rules +
statistics



ML anomaly signal
Isolation Forest



Relationships
NetworkX graph



Scoring
0-100 priority



AI Explanation
LLM summary

DATA & BACKEND



FastAPI backend
+ pandas / openpyxl



Supabase PostgreSQL
vendors · tenders · alerts

FEASIBILITY



TEAM READINESS

- Python already known by the team
- Cuts development & learning time



SCOPED MVP

- Start with CSV/Excel uploads
- Focused set of 4-5 anomaly signals



EXPLAINABLE BY DESIGN

- Rule-based signals work without
- a large labeled fraud dataset
- ML added as a supporting signal only



READY INFRASTRUCTURE

- Supabase: auth + scalable storage
- Modular backend; synthetic → real data



Most importantly: alerts are investigation leads, not accusations – reducing the risk of false conclusions.

BUSINESS STRATEGY



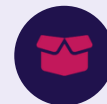
TARGET USERS

Government audit teams, procurement departments, vigilance & oversight bodies



VALUE

Cuts manual review effort; focuses investigators on the cases that matter most



CORE PRODUCT

Procurement monitoring dashboard + anomaly alerts + evidence-based cases



Differentiator: explainable alerts + relationship mapping – not a simple “fraud / no-fraud” prediction

POSSIBLE MODEL

SaaS for departments & agencies, with plans based on procurement volume, users and analysis needs.

FUTURE EXPANSION

Live procurement feeds, document analysis, stronger graph analytics, historical benchmarking, investigator case management.